

PRITHVEX: A Calibrated Uncertainty Platform for Supply Chain Risk Intelligence

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Abstract

Global supply chains operate under chronic information asymmetry: demand and risk forecasts are typically presented as static point estimates with no indication of how reliable they are. PRITHVEX is a calibrated, multi-modal forecasting and risk-intelligence engine that pairs ensemble machine learning with conformal prediction to produce uncertainty-aware, explainable, and continuously self-correcting risk scores for supply chain decision-makers. This paper presents the model design, the calibration methodology, the explainability layer, and the empirical outcomes observed when PRITHVEX is deployed as the AI core of the ENVIREX risk-intelligence platform.

1. Introduction

Disruptions such as typhoons, port strikes, supplier shutdowns, and geopolitical conflicts regularly invalidate demand forecasts within days. Procurement and supply-chain teams are left reacting to events rather than anticipating them, in part because most forecasting systems give no honest signal of their own confidence. The 2021 Suez Canal blockage, which is estimated to have held up roughly \$9.6 billion in trade per day, is a widely cited example of how quickly cascading disruption can outpace traditional planning tools. Industry surveys repeatedly find that a majority of supply chain leaders (cited at 73% in our problem analysis) say they lack real-time risk visibility, and that the average organisational response lag to a disruption is on the order of three to five days.

The central thesis of this work is that the missing ingredient is not more data, but an intelligent decision-support layer that converts raw uncertainty into calibrated, explainable, and actionable risk intelligence. PRITHVEX was designed to fill exactly that gap.

2. Related Approaches

Conventional SCM forecasting pipelines generally fall into one of two camps. Point-forecast systems (classical statistical models or single neural networks) provide a single demand or risk number with no calibrated interval, which causes planners to either over-trust or ignore the output. Black-box ensemble systems improve raw accuracy but rarely expose why a score was produced, which limits adoption in regulated or high-stakes procurement decisions. PRITHVEX is positioned at the intersection of three established but rarely combined techniques: gradient-boosted and deep sequence ensembles for raw predictive power, split conformal prediction for distribution-free uncertainty guarantees, and SHAP attribution for transparent, auditable explanations.

3. System Overview

PRITHVEX is organised as a five-stage pipeline: data collection and integration, uncertainty calibration and risk scoring, dashboard monitoring and alerting, scenario simulation and recommendation, and a closed

decision-implementation feedback loop. The pipeline sits on three underlying system layers — cloud infrastructure, a Bayesian probabilistic core, and a Monte Carlo simulation engine — that are shared across all dashboard panels.

Stage	Function	Reported Outcome
1. Data collection & integration	IoT sensors, ERP systems, third-party feeds onboarded	Sources onboarded
2. Uncertainty calibration & risk scoring	Bayesian inference, feature engineering, ECE evaluation	Model trained, +30% accuracy
3. Dashboard monitoring & alerting	React frontend, WebSocket streaming, calibrated alert engine	Live heatmaps, ~40% faster response
4. Scenario simulation & recommendations	Monte Carlo sims, what-if branching, auto risk-aware suggestions	Decisions optimised, ~25% lower holding cost
5. Decision implementation & feedback loop	Use cases logged, model recalibrated, continuous learning	Adapting, self-improving model

4. Model Architecture

The predictive core is an ensemble of XGBoost, LightGBM, LSTM, and Transformer models, selected to cover tabular feature interactions, short-horizon temporal patterns, and longer-range anomaly detection respectively. Forecast horizons are produced at 7-day, 30-day, and 90-day windows. Predictions are not consumed directly; they are passed through the forecast uncertainty engine described in Section 5 before being surfaced to any user interface.

5. Calibrated Uncertainty Layer

PRITHVEX uses MAPIE-based split conformal prediction to wrap raw ensemble outputs in confidence-bounded intervals with mathematically provable coverage guarantees, meaning the reported intervals can be independently verified rather than taken on faith. An adaptive KL-divergence monitor continuously checks for distributional shift; when a shift is detected — for example during a black-swan event such as a pandemic or a sudden geopolitical shock — the model automatically queues itself for recalibration on new data rather than silently degrading. A full treatment of this layer is given in the companion methodology and calibration documents.

6. Explainability

Every risk score is paired with a SHAP attribution trail that ranks the contributing risk drivers (for example heavy rainfall, port congestion, supplier shutdown probability, or lead-time variance) and assigns each one a signed contribution. This produces a fully auditable, non-black-box output that satisfies the explainability expectations of regulators and enterprise risk committees alike.

7. Empirical Outcomes

- End-to-end pipeline latency of roughly 3 seconds from data ingestion to recommendation delivery.
- Approximately 30% improvement in model accuracy after calibration relative to the uncalibrated baseline.
- Approximately 40% faster disruption response time via the live alerting and heatmap panels.
- Approximately 25% reduction in holding cost from scenario-driven, risk-aware recommendations.
- Approximately 34% reduction in projected carbon footprint from calibrated, right-sized safety buffers.

8. Conclusion

By combining ensemble forecasting, conformal calibration, SHAP-based explainability, and a closed feedback loop, PRITHVEX turns raw predictive uncertainty into a trustworthy, actionable decision-support layer for supply chain risk management. Ongoing and future work — including satellite-based monitoring, autonomous agentic recommendations, and federated learning across supplier networks — is summarised in the accompanying executive report.